CAPSTONE PROJECT

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TITLE PROJECT

REGIONAL ANALYSIS OF ELECTRICITY CONSUMPTION AND EFFICIENCY IN INDONESIA: THE IMPACT OF GRDP AND POPULATION ON NATIONAL ELECTRICITY DEMAND USING IBM GRANITE AI (2017–2023)

RAW DATASET LINK

Electricity Distributed by Province (GWh)	Link
Population by Province	Link
Gross Regional Domestic Product (GRDP) by Province	Link

PROJECT OVERVIEW

Electricity is a vital component of economic development and public welfare. Along with economic and population growth, national electricity demand in Indonesia has shown a consistent upward trend over the years.

This project analyzes interprovincial electricity consumption in Indonesia from 2017 to 2023, focusing on the influence of Gross Regional Domestic Product (GRDP) and population size on national electricity demand.

In addition to traditional statistical and visualization methods, the project integrates IBM Granite AI to:

1. Classify provinces based on energy efficiency levels (Low, Medium, High).
2. Generate automated insights and policy recommendations for sustainable national energy planning.

The analysis provides valuable information for PLN (the State Electricity Company), the Ministry of Energy and Mineral Resources (ESDM), and regional governments in developing data-driven, equitable, and efficient energy distribution strategies across Indonesia.

ANALYSIS PROCESS

1 Setup and Library Installation

The analysis used Python with pandas, matplotlib, seaborn, scikit-learn, and IBM Granite via langchain_community to prepare the environment for data processing and AI interpretation.

2 Data Import and Initial Check

The dataset was imported and examined to confirm its structure, variable types, and absence of missing values.

3 National Electricity Consumption Trend Analysis

The trend of national electricity consumption was visualized using line charts and dual-axis plots to show the relationship between electricity usage and GRDP growth.

4 Electricity Consumption and GRDP per Capita Analysis

Derived variables of electricity consumption per capita and GRDP per capita were created to assess energy efficiency across provinces, and the results were visualized using horizontal bar charts.

5 Energy Efficiency Calculation

Energy efficiency was calculated as the ratio between GRDP per capita and electricity consumption per capita, providing an efficiency index for each province.

6 Efficiency Classification (AI Granite & Quantile)

Both AI-based and statistical quantile classifications were compared to evaluate consistency.

ANALYSIS PROCESS

7

National and Regional Visualization Analysis

Consumption patterns were explored through bar, grid, grouped, pie, scatter, and heatmap visuals.

8

Visualization Summary

Visual insights were synthesized into a narrative for AI Granite interpretation.

9

Multiple Linear Regression (OLS)

OLS measured the effect of GRDP and population on electricity consumption, supported by regression visuals.

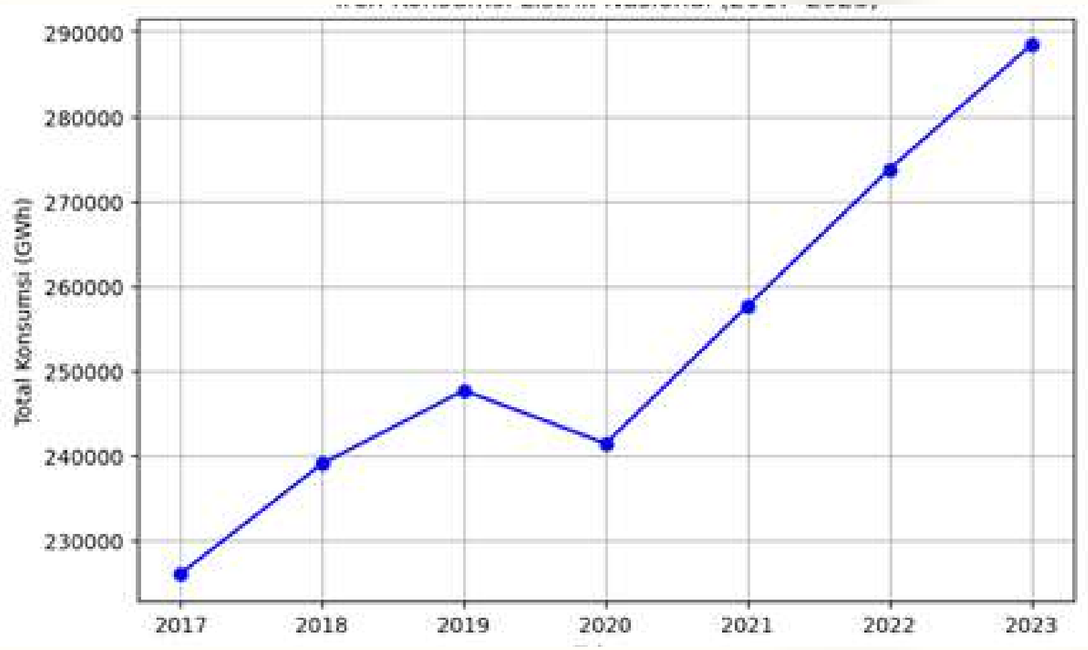
10

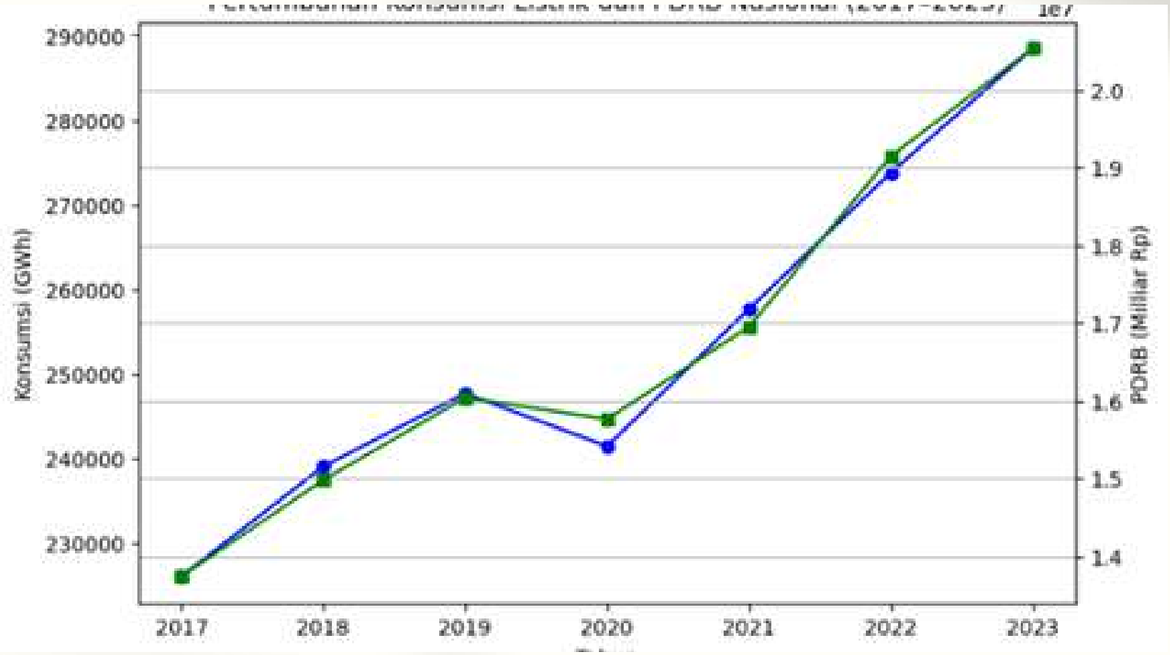
Regression Summary and AI Interpretation

AI Granite provided deeper interpretation and national policy recommendations based on regression outcomes.

INSIGHT & FINDINGS (AND VISUALIZATION)

1 National electricity consumption trends





National electricity consumption showed steady growth from 2017 to 2023, in line with national GDP growth. This indicates that energy demand continues to increase in tandem with the expansion of economic activity across Indonesia.

INSIGHT & FINDINGS (AND VISUALIZATION)

2 Interprovincial Energy Efficiency Analysis

=== Comparison of AI vs Quantile Classification ===

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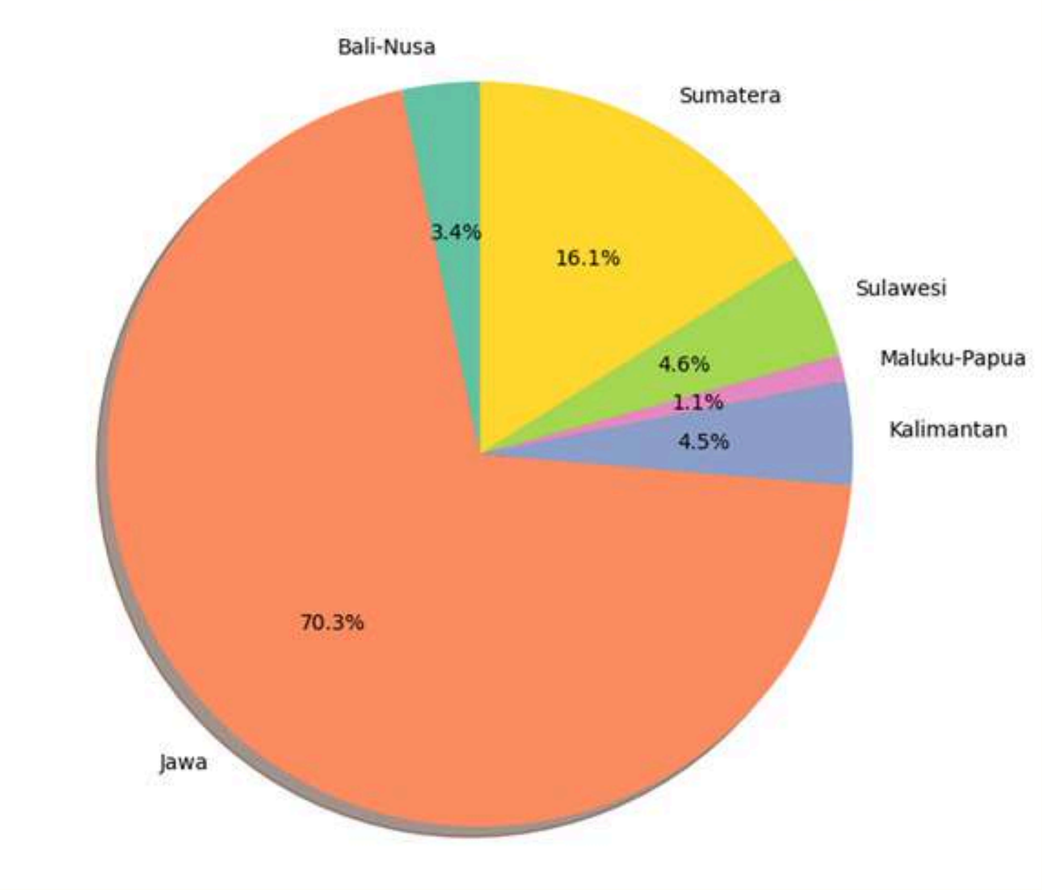
index	Provinsi	Kategori Efisiensi (AI)	Kategori Efisiensi (Quantile)
0	Aceh	Efisiensi Rendah	Efisiensi Rendah
1	Bali	Efisiensi Rendah	Efisiensi Rendah
2	Banten	Efisiensi Rendah	Efisiensi Rendah
3	Bengkulu	Efisiensi Sedang	Efisiensi Sedang
4	DI Yogyakarta	Efisiensi Sedang	Efisiensi Rendah
5	DKI Jakarta	Efisiensi Tinggi	Efisiensi Sedang
6	Gorontalo	Efisiensi Sedang	Efisiensi Sedang
7	Jambi	Efisiensi Tinggi	Efisiensi Tinggi
8	Jawa Barat	Efisiensi Rendah	Efisiensi Rendah
9	Jawa Tengah	Efisiensi Sedang	Efisiensi Rendah
10	Jawa Timur	Efisiensi Sedang	Efisiensi Rendah
11	Kalimantan Barat	Efisiensi Tinggi	Efisiensi Sedang
12	Kalimantan Selatan	Efisiensi Sedang	Efisiensi Rendah
13	Kalimantan Tengah	Efisiensi Tinggi	Efisiensi Tinggi
14	Kalimantan Timur	Efisiensi Tinggi	Efisiensi Tinggi
15	Kalimantan Utara	Efisiensi Tinggi	Efisiensi Tinggi
16	Kepulauan Bangka Belitung	Efisiensi Sedang	Efisiensi Rendah
17	Kepulauan Riau	Efisiensi Tinggi	Efisiensi Tinggi
18	Lampung	Efisiensi Sedang	Efisiensi Sedang
19	Maluku	Efisiensi Sedang	Efisiensi Sedang
20	Maluku Utara	Efisiensi Sedang	Efisiensi Sedang
21	Nusa Tenggara Barat	Efisiensi Sedang	Efisiensi Rendah
22	Nusa Tenggara Timur	Efisiensi Tinggi	Efisiensi Tinggi
23	Papua	Efisiensi Tinggi	Efisiensi Tinggi
24	Papua Barat	Efisiensi Tinggi	Efisiensi Tinggi
25	Riau	Efisiensi Tinggi	Efisiensi Tinggi
26	Sulawesi Barat	Efisiensi Sedang	Efisiensi Tinggi
27	Sulawesi Selatan	Efisiensi Sedang	Efisiensi Sedang

The AI Granite classification shows a pattern consistent with the quantile method, indicating alignment between statistical and AI-based approaches. Eastern provinces such as Papua and East Nusa Tenggara demonstrate high efficiency due to a balanced relationship between economic growth and electricity consumption, while Java-Bali regions tend to have lower efficiency because of high electricity usage that is not proportional to their energy productivity. Overall, interprovincial energy efficiency is influenced not only by economic level but also by consumption patterns and energy management practices within each region.

INSIGHT & FINDINGS (AND VISUALIZATION)

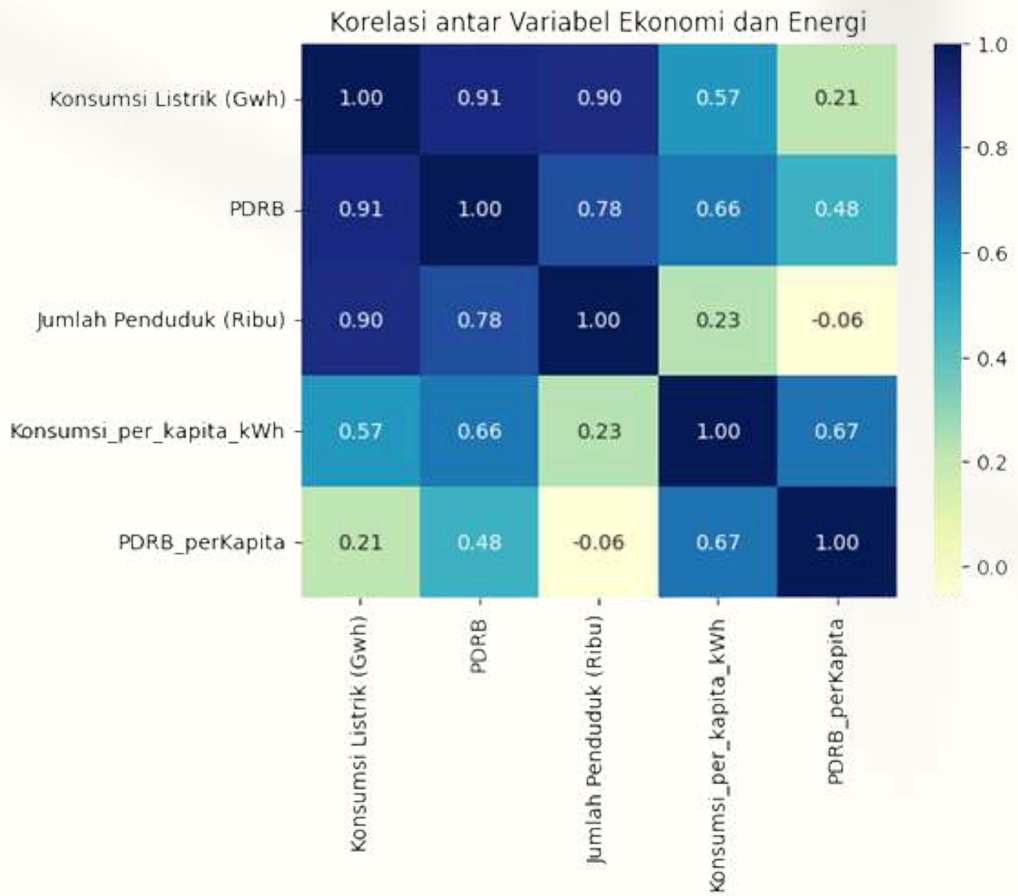
3 Regional Electricity Consumption Comparison



Java Island dominates national electricity consumption (>70%), followed by Sumatra (~16%), while Kalimantan, Sulawesi, and eastern regions contribute smaller shares.
This imbalance reflects regional disparities in energy use, which could affect national energy resilience.

INSIGHT & FINDINGS (AND VISUALIZATION)

4 Correlation and Relationship Analysis



Correlation analysis indicates a strong relationship between electricity consumption and:

- GRDP (r = 0.91)
- Population (r = 0.90)

Meanwhile, the GRDP per capita–electricity per capita relationship (r = 0.57) shows a moderate correlation, meaning economic activity influences electricity consumption more strongly than population size.

INSIGHT & FINDINGS (AND VISUALIZATION)

5 Multiple Linear Regression – Drivers of Electricity Consumption

Variable	Coefficient	p-Value	Interpretation
GRDP (trillion Rp)	0.0093	0	An increase of 1 trillion rupiah in GRDP raises electricity consumption by 0.0093 GWh.
Population (thousand people)	0.5521	0	An increase of 1,000 people raises electricity consumption by 0.5521 GWh.

The R^2 value of 0.927 indicates that 92.7% of the variation in electricity consumption is explained by GRDP and population.

Both variables have a positive and significant effect, with GRDP showing a slightly stronger influence, meaning that economic activity plays a major role in increasing electricity demand in Indonesia.

CONCLUSION

Overall, the results indicate that the increase in Indonesia's national electricity consumption from 2017 to 2023 is strongly influenced by economic growth and population size, with the economic factor playing a more dominant role. Java Island records the highest electricity consumption, while energy efficiency varies across other regions, reflecting disparities between provinces. The analysis also shows that economically advanced regions are not always more energy-efficient. Insights generated by the AI Granite model support these findings and highlight the importance of policies that focus on equitable energy consumption, improved efficiency, and the development of renewable energy sources to achieve sustainable national energy security.

RECOMMENDATION

Based on the findings and AI Granite insights, several strategic recommendations can be proposed. First, the government should promote balanced regional energy development by improving infrastructure and expanding access to electricity outside Java to reduce regional disparities. Second, energy efficiency programs must be strengthened, especially in provinces with high consumption levels, through incentives for energy-saving technologies and industrial optimization. Third, investment in renewable energy sources should be prioritized to diversify the national energy mix and reduce reliance on fossil fuels. Fourth, continuous monitoring and data-driven policy evaluation are essential to ensure effective implementation and sustainable energy management. Lastly, collaboration between regions should be encouraged to exchange best practices and jointly achieve equitable and efficient national energy use.

AI SUPPORT EXPLANATION

1. Energy Efficiency Classification

IBM Granite automatically classified provinces into low, medium, and high efficiency levels using the ratio of GRDP per capita to electricity consumption per capita.

2 Strategic and Policy Analysis

The model provided data-driven policy recommendations for improving regional efficiency and promoting balanced electricity distribution across provinces.

3. AI-Generated Summary and Statistical Interpretation

AI Granite summarized key national visualization and regression findings, highlighting major trends in electricity demand, economic growth, and population dynamics to ensure the insights were analytically sound and contextually relevant.

THANK YOU